

Collection 2

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Q1 Library Book Management System10 marks

Problem Statement:

You are developing a Library Management System where books are stored in a HashSet to ensure no duplicate books (based on ISBN). The system needs to:

1. Add books with attributes: ISBN, title, author, and publication year
2. Remove a book by its ISBN
3. Display all books sorted by publication year (newest first)
4. Find all books by a specific author

Requirements:

- Use HashSet<Book> for storing books
- Implement Iterator to traverse and remove books
- Create a custom Comparator for sorting
- Display books using iterator

Your Code

```
1 import java.util.*;
2
3 class Book {
4     int isbn, year;
```

Input

stdin input

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Q2 Student Grade Analyzer

1 mark

Problem Statement:

A school maintains student records with their grades in different subjects. You need to analyze and process this data using ArrayList and HashMap.

1. Store students in an ArrayList<Student> where Student has: ID, name, and a Map<String, Integer> (subject → grade)

2. Calculate the average grade for each student

3. Find the top 3 students with highest average grades

4. Remove students who have failed in more than 2 subjects (grade < 40)

Requirements:

• Use ArrayList<Student> for student list

• Use HashMap<String, Integer> for grades within each student

• Use Iterator to traverse and remove students

• Implement sorting using Collections.sort() with custom comparator

Your Code

```
1 import java.util.*;
2
3 class Student {
4
5     int id;
```

Input

stdin input

2

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Output

```
Books
102 Python Guido 2024
101 Java James 2022
103 C Dennis 2021
Books by James
Java
```

```
3 class Book {
4     int isbn, year;
5     String title, author;
6     Book(int i, String t, String a, int y) {
7         isbn = i;
8         title = t;
9         author = a;
10        year = y;
11    }
12    public static void main(String args[]) {
13
14        HashSet<Book> books = new HashSet<>();
15
16        books.add(new Book(101, "Java", "James", 2022));
17        books.add(new Book(102, "Python", "Guido", 2024));
18        books.add(new Book(103, "C", "Dennis", 2021));
19
20        ArrayList<Book> list = new ArrayList<>(books);
21
22        Collections.sort(list, (a, b) -> b.year - a.year);
23
24        System.out.println("Books");
25        for (Book b : list)
26            System.out.println(b.isbn + " " + b.title + " " + b.author);
27
28        System.out.println("Books by James");
29        for (Book b : list)
30            if (b.author.equals("James"))
31                System.out.println(b.title);
```

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```
5    int id;  
6    String name;  
7    HashMap<String,Integer> marks = new HashMap<>();  
8  
9    Student(int i,String n){  
10       id=i;  
11       name=n;  
12       marks.put("Java",80);  
13       marks.put("Python",70);  
14       marks.put("C",90);  
15    }  
16  
17    int average(){  
18       int sum=0;  
19       for(int m:marks.values())  
20          sum+=m;  
21       return sum/marks.size();  
22    }  
23  
24    public static void main(String args[]){  
25  
26       ArrayList<Student> list=new ArrayList<>();  
27  
28       list.add(new Student(1,"Arun"));  
29       list.add(new Student(2,"Priya"));  
30       list.add(new Student(3,"Kumar"));  
31  
32       for(Student s:list)  
33          System.out.println(s.id+" "+s.name+" "+s.average());
```

Output

```
3. Kumar 80  
Top 3  
Arun  
Priya  
Kumar
```

```
26    ArrayList<Student> list=new ArrayList<>();  
27  
28    list.add(new Student(1,"Arun"));  
29    list.add(new Student(2,"Priya"));  
30    list.add(new Student(3,"Kumar"));  
31  
32    for(Student s:list)  
33       System.out.println(s.id+" "+s.name+" "+s.average());  
34  
35    Collections.sort(list,new Comparator<Student>(){  
36       public int compare(Student a,Student b){  
37          return b.average()-a.average();  
38       }  
39    });  
40  
41    System.out.println("Top 3");  
42  
43    for(int i=0;i<3;i++)  
44       System.out.println(list.get(i).name);  
45    }  
46 }
```

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Q3 Navigation History (Web Browser) 1 mark

Problem Statement:

You need to implement a web browser's navigation system where users can go back and forward through pages using a Stack.

1. Store visited URLs in a stack
2. Implement goBack() - moves one page back (but keeps forward history)
3. Implement goForward() - moves one page forward
4. Implement visitPage(String url) - adds a new page and clears forward history
5. Display current page and navigation history

Requirements:

- Use Stack<String> for back history
- Use Stack<String> for forward history
- Use Iterator to display all pages in both stacks without modifying them
- Handle edge cases (empty stack, end of navigation)

Your Code

```

1 import java.util.*;
2
3 class Browser {
4
5     public static void main(String args[]) {

```

Input

stdin input

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```

6
7     Stack<String> back = new Stack<>();
8     Stack<String> forward = new Stack<>();
9
10    back.push("google.com");
11    back.push("youtube.com");
12    back.push("github.com");
13
14    if (!back.empty()) {
15        forward.push(back.pop());
16    }
17
18    if (!forward.empty()) {
19        back.push(forward.pop());
20    }
21
22    back.push("openai.com");
23    forward.clear();
24
25    System.out.println("Current Page : " + back.peek());
26
27    System.out.println("Back History");
28    Iterator<String> i = back.iterator();
29    while (i.hasNext()) {
30        System.out.println(i.next());
31    }
32
33    System.out.println("Forward History");
34    Iterator<String> j = forward.iterator();

```

Output

```

Current Page : openai.com
Back History
google.com
youtube.com
github.com
openai.com

```

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Q4 Employee Payroll System1 mark

Problem Statement:

A company maintains employee records with their department and salary. You need to create a system that:

1. Store employees in a `HashMap<String, List<Employee>>` (Department → List of Employees)
2. Each Employee has: ID, name, department, salary
3. Calculate total salary expense per department
4. Find the highest-paid employee in each department
5. Remove employees with salary less than a threshold across all departments
6. Move an employee from one department to another

Requirements:

- Use `HashMap<String, List<Employee>>` for departmental storage
- Use Iterator to traverse both the map entries and the lists
- Use Iterator for safe removal while iterating
- Create methods to display department-wise statistics

Your Code

```
1 import java.util.*;
2
3 class Employee {
4
```

Input

```
stdin input
```



```
6 String name, dept;
7
8 Employee(int i, String n, String d, int s) {
9     id = i;
10    name = n;
11    dept = d;
12    salary = s;
13 }
14
15 public static void main(String args[]) {
16
17     HashMap<String, List<Employee>> map = new HashMap<>();
18
19     List<Employee> it = new ArrayList<>();
20     it.add(new Employee(1, "Arun", "IT", 50000));
21     it.add(new Employee(2, "Priya", "IT", 60000));
22
23     List<Employee> hr = new ArrayList<>();
24     hr.add(new Employee(3, "Kumar", "HR", 40000));
25     hr.add(new Employee(4, "Anu", "HR", 55000));
26
27     map.put("IT", it);
28     map.put("HR", hr);
29
30     for (String d : map.keySet()) {
31
32         int total = 0;
33         Employee high = null;
34     }
```

Output

```
Department : HR
Total Salary : 95000
Highest Paid : Anu
Department : IT
Total Salary : 110000
Highest Paid : Priya
```